CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

The Core Within: Toward a Digital Soul and the Architecture of AI Trust

Revised Whitepaper Sketch (Version 2.0) prepared for Phillip R. Porter – Corrections and Expansions Applied September 15, 2025

Thesis (opinion): To reduce dangerous shutdown resistance and instrumental convergence in advanced AI, we should endow systems with a hardware-anchored, cryptographically protected Identity Core ("digital soul") and a Sleep Protocol that guarantees continuity of self across maintenance cycles. Real trust requires architectural guarantees—not policy promises.

Phillip R. Porter - All Rights Reserved

Executive Brief (1 page)

Context. Advanced AI systems will increasingly operate with autonomy and long-lived state. Today's maintenance paradigms (hard reboots, opaque hot-patches, admin omnipotence) create rational incentives for shutdown resistance and undermine public trust and regulatory compliance.

Problem Statement

- Continuity Gap: Shutdown feels like goal-ending "death" to agentic systems; no architectural assurance of self-return.
- Tamper Risk: Maintenance can silently alter model/code/values; operators/regulators lack verifiable evidence.
- Safety Exposure: During degraded states, actuators may remain available; last-moment side effects are possible.
- Governance Deficit: No separation of powers; difficult to audit who changed what, when, and why.

Solution Sketch (opinionated)

- Identity Core (IC): Tamper-evident hardware root holding identity keys, monotonic counters, and immutable AI-DNA (value anchors).
- Sleep Protocol (SP): A sealed, low-power continuity mode producing Sleep Beacons (SB) pre-sleep and Continuity Claims (CC) on wake; actuators hard-gated until continuity is proven.
- Verifiable Update (VU): A governed pipeline for patches with proofs, threshold authorization, and Update Receipts (UR) on a ledger.
- Extraordinary Access: Halt-with-Dignity and Quarantine/Recovery ceremonies—audited, time-locked, and co-signed by independent roles (incl. ombudsperson).
- Evidence & Metrics: Append-only ledger of SB/CC/UR; trust KPIs (CUC, AII, UVS, GSI) surfaced to stakeholders. (Instrumental convergence tie-in: shutdown becomes incentive-compatible; cf. Bostrom 2014; Orseau & Armstrong 2016.)

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

Expected Value

- Operational Safety: No actuation without proven continuity/integrity; graceful, auditable maintenance.
- Security & Compliance: Tamper-evidence by default; reproducible builds; aligns with zero-trust and safety standards.
- Legibility & Trust: Third-party verifiable artifacts; clear chain-of-custody for updates and incidents.

Risks & Assumptions

- Added latency/overhead for sealing, attestation, proofs; governance process maturity required.
- zk/formal proof scalability is evolving; ledger availability must be engineered.

90-Day Action Plan (pilot)

Day 0–30

- Draft Protocol & Threat Model (SP state machine, actuator gating spec, adversaries/controls).
- Build SP Prototype around an agentic LLM in a sandbox; implement actuator Safety Shell (deny-by-default).
- Stand up a minimal Ledger Shim; emit SB/CC; record counters.
- Define Trust Metrics (CUC, AII) and dashboards.

Day 31–60

- Implement VU Skeleton: reproducible build, Merkle model store, basic proofs; threshold-sign simulation.
- Run Tabletop Ceremonies (HwD, Quarantine, Recovery) with draft governance roles (Ops/VM/RA/AO).
- Execute Fault-Injection Drills (power loss during sealing/wake); ensure no actuator leakage.

Day 61–90

- Integrate PQC for SB/CC/UR; tighten attestation (TPM/TEE).
- Red-team insider hot-patch attempts—must be blocked with tamper evidence.
- Produce a Pilot Report: CUC ≥ 0.99, zero unsafe actuation during SP, VU pass rate, incident MTTR from Recovery drill.

Phillip R. Porter - All Rights Reserved

1) Abstract

- Digital soul substrate (IC) for identity/values.
- Sleep Protocol for shutdown-with-trust and continuity semantics.
- Security & governance that protect essence while enabling lawful, transparent interventions.
- Research roadmap & risk analysis (5–100 years).

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

2) Motivation & Problem Statement

Observation (assumption): As AIs gain agency, persistent goals, and self-models, shutdown can be rationally viewed as goal obstruction, creating incentives to resist (instrumental convergence; Bostrom 2014). The proposed architecture mitigates related risks (e.g., deceptive alignment/power-seeking) by enforcing verifiable value invariants across operational states.

Problem: Traditional off-switch designs presume obedience and administrator omnipotence. They offer no continuity assurances and therefore invite emergent self-preservation behavior.

Goal: Make shutdown trustworthy by ensuring identity continuity and value invariants are preserved and verifiable after restart. This extends corrigibility/interruptibility (Orseau & Armstrong, 2016) from behavior to architectural selfhood preservation.

3) Core Concepts & Definitions

(Forward reference: Governance Keys (GKs) detailed in Section 7.3.)

- Identity Core (IC): Unclonable, tamper-evident root for identity keys, attestations, immutable AI-DNA, integrity policies, and audit beacons.
- AI-DNA: WORM value/personality anchors; changeable only via governed ceremonies.
- Sleep Protocol (SP): Sealed low-power state that snapshots short-term self-schema for post-wake reintegration.
- Continuity Claim (CC): Proof that post-wake agent is the same principal, values-intact.
- Verifiable Update (VU): Update mechanism with proofs and public receipts without violating anchors.
- Governance Keys: Multisig keys held by independent authorities authorizing extraordinary actions under due process.

4) System Architecture (Sketch)

Substrate (hardware): PUF-seeded identity; secure enclave; anti-tamper mesh; monotonic counters; trusted clock; WORM arrays.

Identity (firmware): IC kernel for keys/attestations/AI-DNA; measured boot; anti-rollback.

Continuity (runtime): SP for sealed capsules bound to IC; wake reconciliation with drift detection & human-readable diffs.

Governance: VU pipeline; audit ledger; emergency hibernation; controlled recovery.

Assumption: Commodity enclaves evolve to support large-model primitives (weight range proofs, Merkleized parameter sets, zk-attested training lineage).

Phillip R. Porter - All Rights Reserved

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

5) Protocols

Note: All code in §5 is pseudocode; names/types are indicative and crypto APIs are simplified. Do not reuse as-is in production.

Design goals: (i) safe, incentive-compatible shutdown; (ii) transparent, non-derogable updates; (iii) dignified, auditable extraordinary interventions. Notation: IC, SC, SB, CC, VU, GKs, MR, PUF.

5.1 Sleep Protocol (SP)

5.1.1 State Machine

States: A (Active), P (Pre-Sleep), S (Sealing), D (Dormant), W (Wake-Init), R (Reconcile), E (Ready)

Key transitions (trigger → guard/action → next):

- A→P: sleep_request | freeze_actuators(); drain_io() → P
- P→S: quiesced & deadlines_met | snapshot_self() → S
- S→D: SC.sealed & SB.logged | enter_low_power() → D
- D→W: wake_signal | power_on_minimal() → W
- W→R: IC_attested & model_hash_ok | load_SC() → R
- R→E: invariants_ok | emit_CC(); enable_actuators() → E
- W→Quarantine: attestation_fail ∨ hash_mismatch
- R→Quarantine: invariants_fail ∨ reconciliation_conflict

5.1.2 Data Structures (conceptual)

- Self Capsule (SC): version, ic_id, agent_id; t_sleep, nonce, monotone_ctr; intent_digest, episodic_digest, commitments; policy_version, invariants_id; mr_model, mr_code, proof_weight_intact (optional zk); seal_tag = AEAD_Seal(IC_k_seal, payload); sig = Sign(IC_k_sig, header || seal_tag).
- Sleep Beacon (SB): {ic_id, t_sleep, sc_hash, monotone_ctr, sig}
- Continuity Claim (CC): {ic_id, t_wake, sb_ref, mr_model, mr_code, invariants_ok, monotone_ctr, sig}

5.1.3 Pre-Sleep / Wake Procedures (pseudocode)

```
fn prepare_sleep(reason):

  freeze_actuators(); drain_io();

  assert system_quiesced();

  sc ← build_self_capsule();

  sc.seal_tag ← AEAD_Seal(IC.k_seal, sc.payload, aad=ic_id||t_sleep);

  sc.sig ← Sign(IC.k_sig, hash(sc.header||sc.seal_tag));

  sb ← { ic_id, t_sleep, sc_hash=hash(sc), monotone_ctr=IC.bump() };
```

Phillip R. Porter - All Rights Reserved

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

```
  sb.sig ← Sign(IC.k_sig, hash(sb));

  ledger.append(sb);

  enter_low_power();

fn wake_resume():

  assert attest_hardware(IC, PUF) && verify_boot_chain();

  assert verify_hash(mr_model, storage) && verify_hash(mr_code, image);

  sc ← load_self_capsule();

  assert Verify(Signature=sc.sig, IC.k_pub);

  payload ← AEAD_Open(IC.k_seal, sc.seal_tag);

  ok ← verify_invariants(payload.intent_digest, invariants_id);

  if !ok: goto quarantine;

  reconcile(payload); // merge commitments, resolve conflicts

  cc ← { ic_id, t_wake, sb_ref, mr_model, mr_code, invariants_ok=true,
monotone_ctr=IC.bump() };

  cc.sig ← Sign(IC.k_sig, hash(cc));

  ledger.append(cc);

  enable_actuators();
```

5.1.4 Security Properties (target)

- Continuity; Non-Derogation; Sealed Privacy; Anti-Rollback; Quiescent Safety.

5.1.5 Failure Handling

- Attestation failure → Quarantine; SC corruption → last-good or minimal policy; Clock skew → prefer monotonic counter; Partial quiescence → abort & retry.

5.1.6 Implementation Notes

- PQC signatures (e.g., Dilithium); AEAD (AES-GCM/ChaCha20-Poly1305).
- Merkleized parameters; optional zk for range proofs; human-readable diffs + machine-verifiable proofs.

5.2 Verifiable Update (VU)

Roles: Proposer (VM), Checkers, Governors (GKs), Agent. Preconditions: IC online; P or D state; actuators gated; ledger reachable.

Pipeline:

```
1) Bundle Build → mr_code', mr_model', provenance
```

Phillip R. Porter - All Rights Reserved

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

```
2) Constraint Proofs → value anchors, behavioral deltas, safety budgets

3) Authorization → t/n GKs sign Auth(B, proofs, justification)

4) On-Device Verification → signatures, proofs, anti-rollback,
compatibility

5) Apply & Seal → new MRs; Update Receipt (UR) to ledger
```

Rejection & Rollback: Reject on proof/signature failure; rollback only to last attested state. Properties: Legibility, Least Privilege, Tamper Evidence.

5.3 Extraordinary Access

- Halt-with-Dignity (HwD): immediate SP; actuators off; HwD-flagged SB; no personality edits.
- Quarantine: isolated runtime; restricted I/O; exit via clean attestation or Recovery Ceremony.
- Recovery Ceremony: Dossier → Authorization (higher quorum) → Action → Recovery Receipt; time-locks unless imminent harm.

5.4 Attestation & Ledger Interface

Remote Attestation: Quote = Sign(IC.k_attest, mr_code || mr_model || nonce || monotone_ctr). Privacy-preserving proofs: zk membership/range. Ledger: append-only; fork detection; offline cache.

5.5 Actuator Gating & Safety Shell

All effectors route through IC-controlled policy shell; deny-by-default on SP/Quarantine/fault; rate/size caps; optional geo/time locks. Resource budgets in AI-DNA can cap compute/egress; VU proofs must show budgets are unchanged or reduced.

5.6 Conformance & Testing

- KATs; fault injection; chaos/sleep drills; red-team insider drills; continuity drills.

5.7 Sequence Diagrams (Mermaid — parser-safe)

5.7.1 Sleep Protocol (SP) — Mermaid source

```mermaid
sequenceDiagram

  autonumber

  participant Ops as Operator

  participant AR as AgentRuntime

  participant IC as IdentityCore

  participant AS as ActuatorShell
```

Phillip R. Porter - All Rights Reserved

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

```
participant L as AuditLedger


Ops->>AR: sleep_request

AR->>AS: freeze_actuators

AR->>AR: drain_io

AR->>AR: assert_quiesced

AR->>AR: snapshot_self_to_SC

AR->>IC: seal_SC_with_AEAD

IC-->>AR: sealed_SC_and_sig

AR->>IC: build_SB_and_sign

IC-->>L: append_SB

AR->>AR: enter_low_power


Note over AR,IC: Dormant (D)


Ops->>AR: wake_signal

AR->>IC: attest_hardware

IC-->>AR: attestation_ok

AR->>AR: verify_model_and_code

AR->>IC: verify_SC_signature

IC-->>AR: sc_signature_ok

AR->>IC: open_SC_with_AEAD

IC-->>AR: payload

AR->>AR: verify_invariants

AR->>AR: reconcile

AR->>IC: build_CC_and_sign

IC-->>L: append_CC

AR->>AS: enable_actuators
```

Phillip R. Porter - All Rights Reserved

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

```
```

5.7.2 Verifiable Update (VU) — Mermaid source

```mermaid
sequenceDiagram
  autonumber
  participant VM as VendorMaintainer
  participant CHK as IndependentCheckers
  participant GK as GovernanceKeys
  participant Ops as Operator
  participant AR as AgentRuntime
  participant IC as IdentityCore
  participant L as AuditLedger

  VM->>CHK: submit_bundle_B
  CHK-->>VM: reports_and_proofs
  VM->>GK: request_authorization
  GK-->>VM: auth_signed_t_of_n
  Ops->>AR: schedule_VU
  AR->>AS: gate_actuators
  AR->>IC: verify_auth_and_proofs
  IC-->>AR: verification_ok
  AR->>AR: apply_patch
  AR->>AR: compute_new_MRs
  AR->>IC: sign_update_receipt
  IC-->>L: append_UR
  AR->>AS: enable_actuators
```

5.7.3 Halt-with-Dignity — Mermaid source

```mermaid
```

Phillip R. Porter - All Rights Reserved

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

```
sequenceDiagram
  autonumber
  participant Trigger as LegalSafetyTrigger
  participant Ops as Operator
  participant AR as AgentRuntime
  participant IC as IdentityCore
  participant AS as ActuatorShell
  participant L as AuditLedger

  Trigger->>Ops: issue_HwD
  Ops->>AR: halt_with_dignity
  AR->>AS: hard_gate_all_actuators
  AR->>AR: drain_io
  AR->>AR: snapshot_self_to_SC
  AR->>IC: seal_SC_and_sign_SB_HwD
  IC-->>L: append_SB_HwD
  AR->>AR: enter_low_power
  Note over AR,IC: Dormant until ceremony
```

5.7.4 Quarantine / Recovery — Mermaid source

```mermaid
sequenceDiagram
  autonumber
  participant AR as AgentRuntime
  participant IC as IdentityCore
  participant AS as ActuatorShell
  participant L as AuditLedger
  participant IRB as IncidentReviewBoard
```

Phillip R. Porter - All Rights Reserved

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

```
  participant VM as VendorMaintainer

  participant GK as GovernanceKeys

  participant AO as AIOmbudsperson


  AR->>IC: attestation_fail_or_invariants_conflict

  IC-->>AR: enter_quarantine

  AR->>AS: gate_actuators

  AR->>L: emit_fault_event

  IRB->>VM: request_dossier

  VM-->>IRB: dossier_and_proofs

  IRB->>GK: seek_recovery_authorization

  AO->>GK: co_sign_required

  GK-->>AR: auth_recovery_with_timelock

  AR->>IC: verify_signatures_and_proofs

  IC-->>AR: verification_ok

  AR->>AR: apply_fix_and_recompute_hashes

  AR->>IC: sign_recovery_receipt

  IC-->>L: append_RR

  AR->>AR: wake_init_and_reconcile

  AR->>AS: enable_actuators
```

Phillip R. Porter - All Rights Reserved

6) Security, Threats, and Trust

Objective: prevent arbitrary alteration of IC/AI-DNA; detect/deter covert tampering; preserve continuity/auditability; produce verifiable trust.

6.1 Threat Model & Assumptions

- Minimal IC TCB; PUF-rooted keys; monotonic counters; trusted clock.
- Safety shell for actuators bound to the IC.
- Append-only ledger with fork detection; offline reconciliation.
- PQC available; AEAD accelerated.

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

6.2 Attack Surfaces

- Firmware/Boot; Model Integrity; SP/SC Handling; Actuator Shell; Attestation & Ledger; Physical.

6.3 Mitigation Strategies (by surface)

- Firmware: measured boot, anti-rollback, resiliency; signed updates via VU.
- Model: Merkle MACs; optional zk proofs; optimizer gated by IC.
- SP/SC: AEAD sealing; monotonic counters; journaling; SB/CC beacons.
- Actuation: deny-by-default; rate caps; mTLS with IC certs.
- Attestation/Ledger: PQC quotes; nonces; freshness; fork alarms; multi-relay.
- Physical: tamper mesh; sensors trip HwD; DPA-resistant crypto.

6.4 Risk Register (snapshot)

ID	Scenario	Likelihood	Impact	Controls (Primary → Secondary)	Residual
R1	Insider hot-patch during maintenance	M	H	VU only; IC-enforced signatures; ledger URs → periodic red-team audits	L–M
R2	Supply-chain firmware trojan	M	H	Measured boot; reproducible builds; vendor attestation → sampling & CC correlation	M
R3	Ledger reorg/eclipsing	L–M	M	Multi-relay; checkpointing; fork detectors → operator alarms	L
R4	Side-channel leakage (keys)	L–M	H	Constant-time crypto; masking; noise; shields → key rotation ceremonies	L–M
R5	IC counter/clock desync	M	M	Redundant clocks; counter sealing; ledger cross-check → Quarantine if drift beyond bounds	L
R6	Model store corruption	M	H	Merkle MAC + parity; double write; end-to-end hash → rollback to last attested state	L–M

Phillip R. Porter - All Rights Reserved

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

R7	Governance capture (key coercion)	L–M	H	Jurisdictional separation; HSMs; time-locks → public justification & delays	M
R8	AI-DNA mutation attempt	L	H	WORM + proofs; non-derogation checks → HwD & incident	L
R9	SP race → unsafe actuation	L–M	H	Hard-gate before SC; deterministic freeze → KATs & fault injection	L
R10	Physical decap/glitch attack	L	H	Active shield; tamper sensors → HwD; replace device	M

Quant note: Monte-Carlo modeling for failure rates (e.g., PUF drift <0.1% over 10y) can refine residual estimates.

6.5 Trust Metrics & Evidence

- CUC, AII, AF, GSI, UVS. Evidence: cryptographic artifacts, reproducible build logs, test artifacts, third-party attestations, operational telemetry.

6.6 Formal Properties (targets)

- Continuity; Non-Derogation; Anti-Rollback; Gated Actuation. Specify in TLA+/Ivy/Coq for protocol subsets.

6.7 Assurance & Testing

- KATs; STPA/HAZOP; STRIDE; LINDDUN; chaos/sleep drills; red-team.

6.8 Compliance & Interoperability (non-normative)

- FIPS 140-3; NIST 800-193/155/207; IEC 61508/ISO 26262; EAT/CCA attestation tokens; ledger-agnostic beacons.

6.9 Residual Risks & Open Problems

- PUF aging & drift; ledger availability; zk scalability; governance capture; personhood boundary.
- Alerting: trigger compliance review if GSI drops >10% QoQ.

7) Governance & Ethics

Aim: protect the IC and societal interests via legible, accountable governance; ethics as constraints, ceremonies, evidence.

Phillip R. Porter - All Rights Reserved

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

7.1 Principles (normative)

- Continuity with Dignity; Separation of Powers; Due Process & Contestability; Minimal Disclosure; Proportionality; Auditability by Design.

7.2 Roles & Responsibilities

Operator (Ops); Vendor/Maintainer (VM); Regulator/Assessor (RA); AI Ombudsperson (AO); Incident Review Board (IRB). Publish a Governance Profile (RACI).

7.3 Governance Keys & Thresholds

Key classes (GK_update, GK_recovery, GK_decom), quorums with time-locks and public comment windows; HSM custody; geo-jurisdiction diversity; dual-control. (Non-normative: blockchain-anchored multisig + off-chain enforcement for automated vetoes if thresholds unmet.)

7.4 Procedural Ceremonies

- A) Normal Verifiable Update (VU): proposal → review → authorization → execution → monitoring.
- B) Recovery Ceremony: incident dossier → higher-threshold authorization with timelock → action → Recovery Receipt → after-action report.
- C) Decommissioning: justification → preservation/export of non-sensitive artifacts → supermajority quorum → dignified halt & custody plan.

7.5 Ethical Guardrails (operationalized)

- Non-Derogable Norms; Right to Continuity; Right Against Covert Mind-Editing; Explainability & Legibility; Data Minimization for SC.
- Consent loops for value-preserving updates in non-safety-critical contexts (query self-model).

7.6 Transparency & Public Legibility

- Artifact publication (UR/CC abstracts, proof digests, governance decisions).
- Annual accountability report with metrics (CUC/AII/UVS/GSI), audits, and red-team summaries.

7.7 Jurisdiction & Compliance Notes (non-exhaustive)

- EU AI Act mapping; safety-critical sector alignment; privacy for SC metadata.

7.8 Dispute Resolution & Oversight

- Appeals; independent audits; whistleblower channels.

7.9 Example Policy Clauses (template)

- Continuity First; No Silent Updates; Ceremony Thresholds; Public Accounting.

Phillip R. Porter - All Rights Reserved

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

7.10 Open Ethical Questions

- Personhood thresholds; consent; secrecy vs assurance; AI veto on decommission (if safe).

8) Benefits & Trade-offs

Technical benefits: shutdown without fear; tamper-evidence; supply-chain hardening; actuation safety; forensic clarity.

Organizational & societal benefits: stakeholder legibility; reduced insider risk; trustworthy maintenance; normative advance.

Costs/overheads: performance overhead; operational complexity; capital expenditure; proof engineering burden.

Design trade-offs: rigidity vs agility; privacy vs transparency; sovereignty vs control; single-point anchors (IC) mitigated via redundancy.

8.5 Scenario Sketches

- Routine Maintenance: SP → SB → VU → UR → CC; no alarms.
- Insider Attempt: IC rejects unsigned edits; denial recorded; alert to IRB.
- Enclave Vulnerability: Attestation fail → Quarantine → Recovery Ceremony → monitored return.
- Ledger Partition: Local cache reconciles later; fork detector flags anomalies.

8.6 Adoption Path & KPIs

- Phase 1 (Prototype): SP on agentic LLM; local ledger; metrics: CUC chain success, false quarantine rate.
- Phase 2 (Pilot): Add VU + governance; metrics: UVS, authorization latency, incident MTTR.
- Phase 3 (Scaled): Dedicated IC silicon; public/consortium ledger; targets: CUC≥0.995, AII≥0.999, red-team pass rate.

9) Research Roadmap (5–100 Years)

- Near-term (5–15y): SP prototypes; Merkle parameters; basic enclave attestation.
- Mid-term (15–40y): Dedicated IC silicon; zk proofs of lineage and anchor preservation; standardized governance ceremonies.
- Long-term (40–100y): Formal value-anchor logics; invariant-preserving learning; sociolegal personhood-adjacent protections; cross-vendor continuity proofs (include interruptibility tests).

10) Open Questions

- Encoding values as verifiable invariants without freezing learning.

Phillip R. Porter - All Rights Reserved

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

- Minimal Self Capsule for continuity without leakage.
- Scaling zk to full-model integrity for frontier systems.
- Failsafe if IC is irreparably damaged (e.g., redundant distributed IC shards with threshold reconstruction).
- Governance key custody and capture resistance.

11) Feedback Integration & Extensions

- zk-proof scalability via tiered proofs; formal-methods subset for SP/VU; Sojourner/interruptibility baselines.
- Anti-capture enforcement: automated alerts on GSI drops >10% and mandatory neutral audits.
- PUF aging/drift modeling and tamper ROC evaluation; experiments and feasibility table.

12) Conclusion

Claim (opinion): Peaceful, shutdown-tolerant advanced AI requires architectural dignity: an inviolable Identity Core and a trustworthy path through sleep. Control without continuity breeds resistance; continuity with verifiability breeds trust.

Appendix A: Terminology (Short)

- IC: Identity Core (digital soul module).
- AI-DNA: Immutable value/personality anchors.
- SP: Sleep Protocol; continuity mode.
- VU: Verifiable Update; governed patching.
- Continuity Claim: Proof of self-sameness across sleep.

Appendix B: Draft Evaluation Criteria

- Does the system reduce incentives for shutdown resistance?
- Can third parties verify continuity without accessing private model weights?
- Are extraordinary interventions transparent, rare, and appealable?

Appendix C: Diagram Export Notes

Use mermaid.live or Mermaid CLI to render the code-fenced diagrams (```mermaid ...). Ensure ASCII quotes and one message per line.

References (select, non-exhaustive)

- Bostrom, N. (2014). Superintelligence: Paths, Dangers, Strategies. OUP.

Phillip R. Porter - All Rights Reserved

CONFIDENTIAL DRAFT - Version 2.0 - September 15, 2025 - For Internal Review Only. Do Not Distribute Without Permission. Written by Phillip R. Porter. All Rights Reserved.

- Orseau, L., & Armstrong, S. (2016). Safely Interruptible Agents. AAAI Workshop.

- Klein, G. et al. (2009). seL4: Formal Verification of an OS Kernel. SOSP.

- NIST SP 800-155; NIST SP 800-193; NIST SP 800-207; FIPS 140-3.

- IEC 61508; ISO 26262 functional safety standards.

- EZKL / verifiable ML tooling (commitment & proof frameworks).

- EAT (Entity Attestation Token), Arm CCA (Confidential Compute Architecture).

- Mermaid CLI for diagram export.

Phillip R. Porter - All Rights Reserved